# KRIPAKARA M. N.

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## PROFESSIONAL SUMMARY

Software engineer with internship experience at Samsung R&D and CodeIO. Built an NLP ensemble reaching 98.48% accuracy for voice-command detection and a full-stack platform serving 5,500+ students, cutting administrative effort by ~60%. Skilled in Python, React, Node.js, RAG, and multi-agent LLM systems.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, JavaScript (ES6+), TypeScript, SQL
**Web Development:** React.js, Node.js, Express.js, REST APIs, FastAPI, HTML5, CSS3
**Databases and Tools:** MongoDB, PostgreSQL, Firebase (Auth, Firestore), Git, Docker, AWS, Linux, Cursor
**AI/ML:** LLM Integration, RAG, Multi-Agent Systems, LangGraph, PyTorch, Hugging Face, LangSmith, NLP
**Core CS:** DSA (500+ problems on GeeksforGeeks, NeetCode), OOP, DBMS, OS, Computer Networks

## EXPERIENCE

**Software Development Intern,** CodeIO Bengaluru, India — Mar 2024 – May 2024

- Developed a full-stack exam management platform **(React.js, Node.js, MongoDB)** serving 5,500+ students.
- **Automated exam scheduling, seating allocation, and hall-ticket generation** (PDFKit**,** Nodemailer) with a capacity and batch-aware allocation algorithm, **cutting administrative effort by ~60%.**
- **Implemented role-based access control (RBAC)** separating admin, invigilator, and student workflows.

**Samsung Research Intern,** Samsung R&D Institute India, Bengaluru — Nov 2022 – Aug 2023

- **Improved voice-command detection accuracy to 98.48% (98.46% F1)** with a 3-model NLP ensemble (**ALBERT, Naive Bayes, rule-based classifier)** using PyTorch, outperforming standalone BERT by ~2.5 pts.
- **Optimized inference latency to 56ms (detection) and 24ms (segmentation)** using Pandas-curated datasets and a lightweight 46MB ALBERT-based ensemble, and built a Gradio interface for real-time model validation.

## PROJECTS

**ContractBreaker AI Reliability Testing Tool | Python, pytest | Finalist, OpenFire Hackathon** — Oct 2026

- Engineered an open-source reliability testing tool to expose duplicate payment risks in Python HTTP clients through controlled failure simulation and retry analysis.
- Built deterministic failure verification from execution traces, integrated Alan AI (hackathon sponsor) via a skill-based approval workflow, and validated with **319 tests and 42 benchmark** cases.

**BillaraAI Smart Invoice Generator | React, TypeScript, Firebase (Auth & Firestore)** — Mar 2026

- Shipped a **full-stack AI invoicing platform (Gemini)** that converts text and audio briefs into structured invoices in seconds, cutting creation time by ~65% vs. manual entry.
- Architected a secure backend with **server-side API proxying**, Firestore RBAC rules, and **rate limiting**.
- Developed a real-time dashboard with penalty calculation, email dispatch, and PDF export.

**LiveDataViz Conversational Data Visualization | React.js, JavaScript (ES6+), RAG, BM25** — Dec 2025

- **Reduced chart creation time by ~80%** by developing an **NL2Code** engine that translates user queries into structured Vega-Lite schemas.
- **Built a hybrid RAG** (**Gemini Embeddings + BM25)** that retrieves dataset statistics, categorical values, and sample records to ground LLM-generated charts in the uploaded data.
- **Designed a vector store** with automatic 30-minute TTL cleanup and explicit session deletion.

**TripBuddy AI Multi-Agent Travel Planner | Python, LangSmith, FastAPI, PostgreSQL, Docker** — Jul 2025

- Orchestrated a **5-agent LangGraph system across 9 workflow nodes** for flight, hotel, weather, budgeting, and itinerary planning with supervisor-based routing.
- Integrated **3 live data providers** via **Model Context Protocol (MCP)** over remote HTTP and local stdio, and added a human-in-the-loop approval step for critical booking decisions.
- Containerized the app with a multi-stage **Docker build (non-root user, health check)** and deployed it.

## EDUCATION

**B.M.S. College of Engineering, Bengaluru** — Dec 2020 – Jun 2024
B.E. in Computer Science and Engineering, CGPA: 8.2/10